SMALL POSITIVE SOLUTIONS FOR SINGULAR PRESCRIBED MEAN CURVATURE EQUATIONS WITH CONVECTION

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ABSTRACT. We study a singular prescribed mean curvature Dirichlet problem with a convection term,

$$-\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right)=\frac{\lambda}{u^\alpha}+f(x,u,\nabla u)\quad\text{in }\Omega,\qquad u>0\quad\text{in }\Omega,\qquad u=0\quad\text{on }\partial\Omega,$$

where $0<\alpha<1$ and the continuous perturbation satisfies

$$0\le f(x,t,\xi)\le k_1|t|^{r_1}+k_2|\xi|^{r_2},\qquad r_1,r_2>1.$$

We prove the existence of a positive solution for all sufficiently small $\lambda>0$. The argument is nonvariational and combines a sub–supersolution scheme with a uniformly elliptic modification of the mean curvature operator, an order truncation, and a contraction of the gradient variable in the convection term. A central point is the construction of the supersolution at the natural singular scale $t_\lambda=\lambda^{1/(1+\alpha)}$. This scale also permits the gradient contraction to be removed a posteriori. In particular, the solutions obtained satisfy the quantitative estimate

$$\|u_\lambda\|_{C^1(\overline{\Omega})}\le C\lambda^{1/(1+\alpha)},$$

and hence converge to zero in $C^1(\overline{\Omega})$ as $\lambda\to0^+$. The method is specifically designed to handle simultaneously the singular zero-order term, the gradient dependence, and the non-uniform ellipticity of the prescribed mean curvature operator.

1. INTRODUCTION

We consider the singular Dirichlet problem

$$\begin{cases}-\operatorname{div}\left(\dfrac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right)=\dfrac{\lambda}{u^\alpha}+f(x,u,\nabla u) & \text{in}\quad\Omega,\\ u>0 & \text{in}\quad\Omega,\\ u=0 & \text{on}\quad\partial\Omega,\end{cases}\tag{1}$$

where $\Omega\subset\mathbb{R}^N$, $N\ge2$, is a bounded domain with smooth boundary, $\lambda>0$, $0<\alpha<1$, and $f:\Omega\times\mathbb{R}\times\mathbb{R}^N\to\mathbb{R}$ is continuous and satisfies

$$0\le f(x,t,\xi)\le k_1|t|^{r_1}+k_2|\xi|^{r_2},\qquad k_1,k_2>0,\qquad r_1,r_2>1.\tag{2}$$

The purpose of the paper is to establish a branch of positive solutions which is quantitatively small as $\lambda\to0^+$, while retaining both the singular forcing $u^{-\alpha}$ and the genuine convection dependence on ∇u.

The operator in (1) is the non-parametric prescribed mean curvature operator. Its geometric origin and the classical Dirichlet theory are connected with minimal and capillary surfaces; see, among others, [11, 13, 21, 28]. Existence and multiplicity for nonlinear prescribed mean curvature equations have been studied through several methods, including barrier arguments, topological and variational techniques, and finite-dimensional approximations; see [8, 9, 14, 16, 22, 25]. Because the operator is not uniformly elliptic, a priori height and gradient bounds play a central role in this theory; see, for instance, [23, 27, 29]. More recently, Figueiredo and Rădulescu [10] treated positive solutions under critical exponential growth, illustrating that the mean curvature operator can also be handled in borderline nonlinear regimes. Recent work has further developed the PDE and BV theory

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of non-parametric mean curvature equations with rough data and nonlinear zero-order terms. Giachetti, Oliva and Petitta [4], for instance, obtained bounded solutions for nonlinear mean curvature problems allowing a factor which may be singular at the origin, while Oliva, Petitta and Segura de León [6] studied the regularizing role of absorption terms. Filippucci and Zheng [3] investigated existence and nonexistence phenomena for weighted mean curvature equations, including singular or degenerate weights.

The singular term in (1) places the problem within the broad theory of singular elliptic equations. Such equations require a different treatment from standard superlinear problems because the right-hand side is not defined at the homogeneous boundary value and typically behaves like a negative power of the distance to $\partial\Omega$; the corresponding boundary regularity and gradient estimates go back to [18, 19] and, for quasilinear operators, to [17]. A recent comprehensive account of this theory is given by Oliva and Petitta [5]. In a related direction, Balducci, Oliva and Petitta [2] considered elliptic equations where singular terms and nonlinear gradient contributions compete. These developments emphasize that the simultaneous presence of singularities and first-order terms is not merely a technical perturbation: it affects the natural functional setting, compactness, boundary estimates, and the construction of admissible barriers.

A second difficulty in (1) is the convection term. Since $f(x, u, \nabla u)$ depends explicitly on the gradient, the equation is in general nonvariational. In particular, the usual energy associated with the area functional does not directly encode the full right-hand side. Sub–supersolution methods combined with truncation provide a flexible alternative in this setting. This philosophy has recently proved useful for quasilinear problems with convection and very large nonlinearities. In [7], positive solutions were obtained for elliptic equations with convection and variable exponents without an upper restriction by introducing suitable contracted and truncated auxiliary problems. Related ideas were developed for quasilinear systems in [1]. The present work adapts this mechanism to the non-uniformly elliptic mean curvature operator and, at the same time, incorporates the singular source $\lambda u^{-\alpha}$.

The interaction between these three features, the mean curvature operator, the singular term, and the gradient-dependent nonlinearity, is the main issue addressed here. The small-parameter regime reveals the appropriate scale. Indeed, setting

$$t_\lambda := \lambda^{1/(1+\alpha)}$$

gives the exact balance

$$\lambda(t_\lambda w)^{-\alpha} = t_\lambda w^{-\alpha}.$$

Consequently, a supersolution of size t_λ has its principal elliptic contribution and its singular forcing at the same order. Meanwhile, assumption (2) yields

$$(t_\lambda w)^{r_1} = o(t_\lambda), \qquad |t_\lambda \nabla w|^{r_2} = o(t_\lambda) \qquad (\lambda \to 0^+),$$

precisely because $r_1, r_2 > 1$. Thus these restrictions are not imposed by a Sobolev critical threshold; rather, they express the perturbative character of the zero- and first-order nonlinearities at the natural singular scale. This observation is the key to the construction and also explains why no Sobolev subcritical restriction on r_1 or r_2 is needed in our argument.

Our approach can be summarized as follows. We first replace the mean curvature coefficient outside the small-gradient region by a uniformly elliptic coefficient which agrees with $(1 + |\nabla u|^2)^{-1/2}$ whenever $|\nabla u| < 1$. The purely singular auxiliary equation supplies a positive lower solution. For the upper solution we use an explicit barrier at scale t_λ, which yields an explicit smallness range for the parameter. We then contract the gradient variable in f and truncate the unknown between the ordered lower and upper solutions. The resulting auxiliary problem can be solved within a uniformly elliptic framework. Localization between the barriers is recovered by comparison. Finally, after the rescaling $u = t_\lambda w$, uniform $C^{1,\sigma}$ estimates show that the gradient contraction is inactive for λ small; the auxiliary coefficient therefore coincides with the original mean curvature coefficient. This closes the argument directly for (1).

The resulting conclusion is stronger than a qualitative existence statement in the small-parameter regime: the proof identifies the asymptotically correct size of the constructed solutions,

$$\|u_\lambda\|_{C^1(\overline{\Omega})} = O\Big(\lambda^{1/(1+\alpha)}\Big).$$

In this sense, the present result connects the modern theory of singular mean curvature problems with the truncation/contraction techniques for convection equations. The main contribution is the implementation of this combined strategy for a singular prescribed mean curvature equation with genuine gradient dependence, together with a quantitative small-solution estimate and an explicit parameter regime produced by the supersolution construction.

By a solution of (1) we mean a function $u \in C^1(\overline{\Omega})$, positive in Ω, such that

$$\int_\Omega \frac{\nabla u \cdot \nabla \varphi}{\sqrt{1+|\nabla u|^2}}\,dx = \int_\Omega \left(\frac{\lambda}{u^\alpha} + f(x,u,\nabla u)\right)\varphi\,dx$$

for every $\varphi \in H_0^1(\Omega)$. The restriction $0 < \alpha < 1$ is particularly convenient here because the boundary behaviour of the positive barriers makes the singular term compatible with this weak formulation.

We can now state the main result.

Theorem 1.1. *Assume $0 < \alpha < 1$ and that (2) holds with constant exponents $r_1, r_2 > 1$. Then there exists $\lambda^* > 0$ such that, for every $\lambda \in (0, \lambda^*]$, problem (1) possesses a positive weak solution*

$$u_\lambda \in C^{1,\beta}(\overline{\Omega})$$

for some $\beta \in (0,1)$ independent of λ. Moreover, if

$$t_\lambda := \lambda^{1/(1+\alpha)},$$

then there exists a constant $C > 0$, independent of small λ, such that

$$\|u_\lambda\|_{C^1(\overline{\Omega})} \le C t_\lambda.$$

In particular,

$$\|u_\lambda\|_{C^1(\overline{\Omega})} \longrightarrow 0 \qquad \text{as } \lambda \to 0^+.$$

The paper is organized as follows. Section 2 introduces the uniformly elliptic modification of the mean curvature operator and the auxiliary singular problem. Section 3 constructs ordered sub- and supersolutions, with the supersolution chosen at the scale $\lambda^{1/(1+\alpha)}$. Section 4 introduces the order truncation and the contraction in the gradient variable and establishes existence for the truncated problem. Section 5 derives the uniform gradient estimate needed to remove the contraction. The final section combines these ingredients to prove Theorem 1.1.

2. Auxiliary results

In the course of the proof of Theorem 1.1, we define an auxiliary quasilinear equation

$$\begin{cases} -\operatorname{div}\left(a(|\nabla u|^2)\nabla u\right) = \dfrac{\lambda}{u^\alpha} + f(x,u,\nabla u) & \text{in} \quad \Omega, \\ u > 0 & \text{in} \quad \Omega \\ u = 0 & \text{on} \quad \partial\Omega. \end{cases} \tag{3}$$

The function $a : [0,\infty) \to \mathbb{R}$ is defined by

$$a(v) = \begin{cases} \dfrac{1}{\sqrt{1+v}} & \text{in} \quad [0,1), \\ b(v-2)^2 + c & \text{in} \quad [1,2), \\ c & \text{in} \quad [2,+\infty) \end{cases} \tag{4}$$

with $b = \frac{1}{8\sqrt{2}}$ and $c = \frac{7}{8\sqrt{2}}$. This function a has been used in [14]. Note that a satisfies the structural ellipticity conditions of [20] and gives rise to a uniformly elliptic operator. For bounded right-hand sides the global $C^{1,\sigma}$ theory of [20] is therefore applicable. In the present problem, however, the term $u^{-\alpha}$ is unbounded at the boundary. The precise boundary-singular regularity statement needed below is recorded and justified in Section 5; see Lemma 5.2. In particular, notice that

$$c \le a(s) \le 1, \forall s \in [0,+\infty). \tag{5}$$

Moreover

$$a(t) \in C^1(0,+\infty) \text{ and } a(t) > 0 \text{ for } t > 0. \tag{6}$$

Definition 2.1. *Let us fix a value of $\lambda > 0$. We say that a function $\overline{\psi} \in C^1(\overline{\Omega})$, positive in Ω, is an upper solution of* (3) *if*

$$\overline{\psi}(x) \geq 0 \quad \text{on } \partial\Omega$$

and

$$\int_\Omega a\left(|\nabla\overline{\psi}|^2\right)\nabla\overline{\psi}\cdot\nabla w\,dx \geq \int_\Omega \Big(\frac{\lambda}{\overline{\psi}^\alpha} + f(x,\overline{\psi},\nabla\overline{\psi})\Big)w\,dx$$

for every $w \in H_0^1(\Omega)$, with $w(x) \geq 0$ a.e. in Ω. Similarly, a lower solution $\underline{\psi} \in C^1(\overline{\Omega})$, positive in Ω, is defined by reversing the two inequalities written above.

Define the function

$$\Phi(t) = \int_0^t sa(s^2)ds, \;\; t \in \mathbb{R}. \tag{7}$$

The following result can be obtained by standard computations.

Lemma 2.2. *Let a as defined in* (4)*, then*

$$sa(s^2) \to 0 \text{ as } s \to 0, \tag{8}$$

$$sa(s^2) \to \infty \text{ as } s \to \infty, \tag{9}$$

$$s \mapsto sa(s^2) \text{ is increasing in } (0,\infty), \tag{10}$$

$$\frac{19}{40} \leq \frac{(sa(s^2))'}{a(s^2)} \leq 1, \;\; s > 0, \tag{11}$$

$$\nu_0 := \frac{19}{40\sqrt{2}} \leq (sa(s^2))' = a(s^2) + 2s^2a'(s^2) \leq 1, \;\; s \geq 0. \tag{12}$$

Proof. By the definition of a in (4), properties (8) and (9) are easy.

Now, we will prove (10). By standard computations we can conclude that

$$a'(v) = \begin{cases} -\frac{1}{2(1+v)^{3/2}} & \text{in} \quad [0,1), \\ 2b(v-2) & \text{in} \quad [1,2), \\ 0 & \text{in} \quad [2,+\infty). \end{cases} \tag{13}$$

Hence, a belongs to $C^1([0,+\infty))$.

We will show that

$$(sa(s^2))' = a(s^2) + 2s^2a'(s^2) > 0 \text{ for } s > 0.$$

Assume $s \in (0,1)$. By (4) and (13) we obtain

$$\begin{aligned} a(s^2) + 2s^2a'(s^2) &= a(s^2) + 2s^2\left(\tfrac{-1}{2(1+s^2)^{3/2}}\right) \\ &= \tfrac{1}{(1+s^2)^{1/2}} + 2s^2\left(\tfrac{-1}{2(1+s^2)^{3/2}}\right) \\ &= \tfrac{1}{(1+s^2)^{1/2}}\left(1 - \tfrac{s^2}{1+s^2}\right) = \tfrac{1}{(1+s^2)^{3/2}} \geq \tfrac{1}{2\sqrt{2}} > \nu_0. \end{aligned}$$

For $s \in [1,\sqrt{2})$ put $v = s^2 \in [1,2)$. By (4), (13) and $c = 7b$ we obtain

$$\begin{aligned} a(v) + 2va'(v) &= b(v-2)^2 + c + 4bv(v-2) \\ &= b\left(5v^2 - 12v + 11\right) \\ &= 5b\left(v - \tfrac{6}{5}\right)^2 + \tfrac{19}{5}\,b \\ &\geq \tfrac{19}{5}\,b = \tfrac{19}{40\sqrt{2}} = \nu_0 > 0. \end{aligned} \tag{14}$$

Assume $s \in [\sqrt{2}, \infty)$. Using (4) and (13) we obtain

$$a(s^2) + 2s^2 a'(s^2) = c + 0 = c = \frac{7}{8\sqrt{2}} > \nu_0.$$

Therefore, $s\,a(s^2)$ is increasing in $(0, \infty)$. Since $a' \leq 0$ and $a \leq 1$, we also have $a(s^2) + 2s^2 a'(s^2) \leq a(s^2) \leq 1$, which proves (12).

Next, we prove (11).

By standard computations

$$\frac{(sa(s^2))'}{a(s^2)} = \frac{a(s^2) + 2s^2 a'(s^2)}{a(s^2)} = 1 + \frac{2s^2 a'(s^2)}{a(s^2)}.$$

Assume $s \in (0, 1)$. By (4) and (13) we obtain

$$\begin{aligned} \frac{2s^2 a'(s^2)}{a(s^2)} &= \frac{2s^2 \left(-\frac{1}{2(1+s^2)^{3/2}}\right)}{\frac{1}{(1+s^2)^{1/2}}} \\ &= \frac{-2s^2}{2(1+s^2)^{3/2}} (1+s^2)^{1/2} \\ &= \frac{-s^2}{1+s^2}. \end{aligned}$$

Hence,

$$1 + \frac{2s^2 a'(s^2)}{a(s^2)} = 1 - \frac{s^2}{1+s^2} = \frac{1}{1+s^2} > \frac{1}{2}$$

and moreover

$$1 + \frac{2s^2 a'(s^2)}{a(s^2)} \leq 1.$$

Therefore,

$$\frac{1}{2} < \frac{(sa(s^2))'}{a(s^2)} \leq 1. \tag{15}$$

Assume $s \in [1, \sqrt{2})$, in particular $s^2 - 2 < 0$. Using (4) and (13) we obtain

$$\frac{2s^2 a'(s^2)}{a(s^2)} = \frac{4bs^2(s^2-2)}{b(s^2-2)^2 + c}.$$

Hence,

$$1 + \frac{2s^2 a'(s^2)}{a(s^2)} = 1 + \frac{4bs^2(s^2-2)}{b(s^2-2)^2 + c} < 1.$$

Moreover, since $b(s^2-2)^2 + c \leq b + c = 8b$ for $s^2 \in [1, 2)$, (14) gives

$$1 + \frac{2s^2 a'(s^2)}{a(s^2)} = \frac{a(s^2) + 2s^2 a'(s^2)}{b(s^2-2)^2 + c} \geq \frac{\frac{19}{5}\, b}{8b} = \frac{19}{40}.$$

Then

$$\frac{19}{40} \leq \frac{(sa(s^2))'}{a(s^2)} < 1. \tag{16}$$

Assume $s \in [\sqrt{2}, +\infty)$. By (4) and (13) we obtain

$$\frac{(sa(s^2))'}{a(s^2)} = 1 + \frac{2s^2 a'(s^2)}{a(s^2)} = 1. \tag{17}$$

As a consequence of (15), (16) and (17), we get

$$\frac{19}{40} \leq \frac{(sa(s^2))'}{a(s^2)} \leq 1, \ s > 0.$$

□

Remark 2.3. *The constant $\frac{1}{2}$ cannot replace $\frac{19}{40}$ in* (11)*. Indeed, for $v = s^2 \in [1,2)$ one has $\frac{(sa(s^2))'}{a(s^2)} = \frac{5v^2-12v+11}{v^2-4v+11}$, whose minimum on $[1,2]$ is attained at $v = (11-\sqrt{77})/2 \approx 1.113$ and is approximately $0.4929 < \frac{1}{2}$. Only the positivity of the lower bound is used in the sequel.*

Remark 2.4. *For $\mathcal{A}(\xi) := a(|\xi|^2)\xi$ one has $D\mathcal{A}(\xi) = a(|\xi|^2)I + 2a'(|\xi|^2)\xi \otimes \xi$. Its eigenvalues are $a(|\xi|^2) \in [c,1]$ (in the directions orthogonal to ξ) and $a(|\xi|^2) + 2|\xi|^2 a'(|\xi|^2) \in [\nu_0, 1]$ (in the direction of ξ), by* (5) *and* (12)*. Since $\nu_0 < c$, it follows that*

$$\nu_0|\eta|^2 \leq D\mathcal{A}(\xi)\eta \cdot \eta \leq |\eta|^2, \qquad \xi, \eta \in \mathbb{R}^N, \tag{18}$$

and consequently

$$\big(\mathcal{A}(\xi) - \mathcal{A}(\zeta)\big) \cdot (\xi - \zeta) \geq \nu_0|\xi - \zeta|^2, \qquad \xi, \zeta \in \mathbb{R}^N. \tag{19}$$

The same bounds hold, with the same constants, for $\mathcal{A}_t(\xi) := a(t^2|\xi|^2)\xi$ for every $t > 0$, since $D\mathcal{A}_t(\xi) = D\mathcal{A}(t\xi)$.

An immediate conclusion is the following one.

Lemma 2.5. *Let $\phi(s) = a(s^2)$, properties (8)-(11) of Lemma 2.2 imply*

$$s\phi(s) \to 0 \text{ as } s \to 0, \tag{20}$$

$$s\phi(s) \to \infty \text{ as } s \to \infty, \tag{21}$$

$$s \mapsto s\phi(s) \text{ is strictly increasing in } (0,\infty), \tag{22}$$

$$\frac{19}{40} \leq \frac{(s\phi(s))'}{\phi(s)} \leq 1, \ s > 0. \tag{23}$$

For the reader's convenience we recall some facts on Orlicz–Sobolev spaces; see [26].
By (6) and Lemma 2.5-(23), the Orlicz space $L_\Phi(\Omega)$ coincides with the set (of equivalence classes) of measurable functions $u : \Omega \to \mathbb{R}$ such that

$$\int_\Omega \Phi(|u|)dx < +\infty. \tag{24}$$

The usual norm on $L_\Phi(\Omega)$ is

$$\|u\|_\Phi = \inf\left\{\mu > 0 : \int_\Omega \Phi\left(\frac{u(x)}{\mu}\right)dx \leq 1\right\}$$

called the Luxemburg norm, and the norm on $W^{1,\Phi}(\Omega)$ is

$$\|u\|_{1,\Phi} = \|u\|_\Phi + \sum_{i=1}^N \left\|\frac{\partial u}{\partial x_i}\right\|_\Phi,$$

known as Orlicz-Sobolev norm, while $W_0^{1,\Phi}(\Omega)$ stands for the closure of $C_0^\infty(\Omega)$ with respect to the norm in $W^{1,\Phi}(\Omega)$.
By (5) and (7) we obtain

$$\frac{c}{2}t^2 \leq \Phi(t) \leq \frac{1}{2}t^2, \ \text{ for } \ t \geq 0. \tag{25}$$

Hence,

$$\frac{c}{2\mu^2}\|u\|^2_{H_0^1(\Omega)} \leq \int_\Omega \Phi\left(\frac{|\nabla u|}{\mu}\right)dx \leq \frac{1}{2\mu^2}\|u\|^2_{H_0^1(\Omega)}, \ \ \forall u \in W_0^{1,2}(\Omega), \ \mu > 0. \tag{26}$$

By (25) and the fact that $c < 1$, we obtain

$$\frac{1}{2}t^2 \leq \frac{1}{c}\Phi(t) \leq \Phi\left(\frac{1}{c}t\right), \tag{27}$$

see [15, Lemma 3.2.2].
By (25) and (26)

$$L_\Phi(\Omega) = L^2(\Omega)$$

and

$$W_0^{1,\Phi}(\Omega) = H_0^1(\Omega). \tag{28}$$

Now, $t^2\phi(t) = t\Phi'(t)$ and, by (23), $\frac{19}{40}\Phi'(t) \leq t\Phi''(t) = t\,(t\phi(t))' \leq \Phi'(t)$ for $t > 0$. Hence $\frac{59}{40}\Phi' \leq (t\Phi')' \leq 2\Phi'$, and integrating on $(0, t)$ (note that $t\Phi'(t) \to 0$ as $t \to 0$) we obtain

$$\frac{59}{40} \leq \frac{t^2\phi(t)}{\Phi(t)} \leq 2,\ t > 0. \tag{29}$$

Defining

$$\underline{\phi} := \inf_{t>0} \frac{t^2\phi(t)}{\Phi(t)} \text{ and } \overline{\phi} := \sup_{t>0} \frac{t^2\phi(t)}{\Phi(t)}, \tag{30}$$

we obtain that

$$\frac{59}{40} \leq \underline{\phi} \text{ and } \overline{\phi} \leq 2. \tag{31}$$

Remark 2.6. *By (29), the function $\Phi(t)$ satisfies the Δ_2-condition, that is*

$$\exists k > 0,\ \Phi(2t) \leq k\Phi(t), \quad t > 0.$$

Firstly, we consider the following auxiliary problem

$$\begin{cases} -\operatorname{div}\left(a(|\nabla u|^2)\nabla u\right) = \dfrac{\lambda}{u^\alpha} & \text{in} \quad \Omega, \\ u > 0 & \text{in} \quad \Omega \\ u = 0 & \text{on} \quad \partial\Omega. \end{cases} \tag{32}$$

Lemma 2.7. *For any constant $\lambda > 0$ the problem (32) admits a solution $u_{0,\lambda} \in C^1(\overline{\Omega})$ and $\partial u_{0,\lambda}/\partial\eta < 0$ on $\partial\Omega$.*

Proof. For $\varepsilon > 0$ consider the regularized problem

$$\begin{cases} -\operatorname{div}\big(a(|\nabla u|^2)\nabla u\big) &= \dfrac{\lambda}{(u+\varepsilon)^\alpha} \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega. \end{cases} \tag{33}$$

Since $c \leq a \leq 1$ and $s \mapsto sa(s^2)$ is strictly increasing, the operator

$$A(u) = -\operatorname{div}\big(a(|\nabla u|^2)\nabla u\big)$$

is bounded, coercive, strictly monotone and of type (S_+) from $H_0^1(\Omega)$ into $H^{-1}(\Omega)$. The right-hand side of (33) is bounded. Hence the Browder–Minty theorem gives a unique $u_\varepsilon \in H_0^1(\Omega)$. Testing with u_ε^- gives $u_\varepsilon \geq 0$; by [20], $u_\varepsilon \in C^{1,\sigma}(\overline{\Omega})$, and the strong maximum principle [24] yields $u_\varepsilon > 0$ in Ω.

If $0 < \varepsilon_1 < \varepsilon_2$, comparison gives $u_{\varepsilon_1} \geq u_{\varepsilon_2}$. Fixing one regularized solution as a positive lower barrier and using the standard boundary barrier $d(x) = \operatorname{dist}(x, \partial\Omega)$, one obtains constants $m_1, m_2 > 0$, independent of $\varepsilon \in (0, 1]$, such that

$$m_1 d(x) \leq u_\varepsilon(x) \leq m_2 \qquad \text{in } \Omega.$$

Consequently

$$0 \leq \frac{\lambda}{(u_\varepsilon + \varepsilon)^\alpha} \leq C\, d(x)^{-\alpha}.$$

Because $0 < \alpha < 1$, Hardy's inequality shows that $d^{-\alpha}$ defines an element of $H^{-1}(\Omega)$:

$$\int_\Omega d^{-\alpha}|\varphi| \le \left(\int_\Omega d^{2(1-\alpha)}\right)^{1/2} \left\|\frac{\varphi}{d}\right\|_2 \le C\|\nabla\varphi\|_2.$$

The energy estimate and monotonicity therefore give, along a subsequence, $u_\varepsilon \to u_{0,\lambda}$ strongly in $H_0^1(\Omega)$ and a.e. in Ω. Passing to the limit in (33) yields

$$-\operatorname{div}\big(a(|\nabla u_{0,\lambda}|^2)\nabla u_{0,\lambda}\big) = \lambda u_{0,\lambda}^{-\alpha}.$$

The boundary estimate above persists in the limit. Since the right-hand side is bounded by $Cd^{-\alpha}$ with $0 < \alpha < 1$, the boundary regularity for singular quasilinear equations [17, Theorem B.1] (see also [18, 19] for the semilinear case) gives $u_{0,\lambda} \in C^{1,\sigma}(\overline{\Omega})$ for some $\sigma \in (0,1)$; the strong maximum principle and Hopf boundary lemma [24] give $u_{0,\lambda} > 0$ in Ω and $\partial u_{0,\lambda}/\partial\eta < 0$ on $\partial\Omega$. □

Definition 2.8. *A function $u \in W^{1,\Phi}(\Omega)$, positive in Ω, is a subsolution of (32) if*

$$\int_\Omega \phi(|\nabla u|)\nabla u\nabla\varphi dx \le \int_\Omega \frac{\lambda}{u^\alpha}\varphi dx,\ \forall\varphi \in W_0^{1,\Phi}(\Omega),\ \varphi \ge 0.$$

A function $u \in W^{1,\Phi}(\Omega)$, positive in Ω, is a supersolution of (32) if

$$\int_\Omega \phi(|\nabla u|)\nabla u\nabla\varphi dx \ge \int_\Omega \frac{\lambda}{u^\alpha}\varphi dx,\ \forall\varphi \in W_0^{1,\Phi}(\Omega),\ \varphi \ge 0.$$

A function $u \in W^{1,\Phi}(\Omega)$, positive in Ω, is a solution of (32) if

$$\int_\Omega \phi(|\nabla u|)\nabla u\nabla\varphi dx = \int_\Omega \frac{\lambda}{u^\alpha}\varphi dx,\ \forall\varphi \in W_0^{1,\Phi}(\Omega).$$

The following comparison principle is a simple version, adapted to the present setting, of [12, Theorem 1.2].

Lemma 2.9 (Comparison Principle)**.** *Let $u_1, u_2 \in W^{1,\Phi}(\Omega)$ be a subsolution and a supersolution of (32), respectively, with $u_1, u_2 > 0$ in Ω, $(u_1 - u_2)^+ \in W_0^{1,\Phi}(\Omega)$, and such that $u_i^{-\alpha}(u_1 - u_2)^+ \in L^1(\Omega)$, $i = 1, 2$. Then $u_1 \le u_2$ a.e. in Ω.*

Proof. Testing the two inequalities with $\varphi = (u_1 - u_2)^+ \ge 0$ and subtracting, we obtain

$$\int_{\{u_1>u_2\}} \big(\mathcal{A}(\nabla u_1) - \mathcal{A}(\nabla u_2)\big)\cdot\nabla(u_1 - u_2)\,dx \le \lambda\int_{\{u_1>u_2\}} \big(u_1^{-\alpha} - u_2^{-\alpha}\big)(u_1 - u_2)\,dx \le 0,$$

because $s \mapsto s^{-\alpha}$ is decreasing. By (19), $\nu_0\|\nabla(u_1 - u_2)^+\|_2^2 \le 0$, hence $(u_1 - u_2)^+ = 0$. □

3. Sub–supersolutions for problem (3)

We construct the supersolution at the scale dictated by the singular term. Put

$$t_\lambda := \lambda^{\frac{1}{1+\alpha}}, \qquad \lambda = t_\lambda^{1+\alpha}.$$

Choose $x_0 \in \mathbb{R}^N \setminus \overline{\Omega}$ and set

$$R_\Omega := \max_{x\in\overline{\Omega}} |x - x_0|, \qquad r_\Omega := \min_{x\in\overline{\Omega}} |x - x_0| > 0.$$

Fix $M_0 := R_\Omega^2 + 1$ and $\vartheta(x) := M_0 - |x - x_0|^2$. Then

$$1 \le \vartheta(x) \le B_0 := M_0 - r_\Omega^2 \le M_0, \qquad |\nabla\vartheta(x)| = 2|x - x_0| \le G := 2R_\Omega. \tag{34}$$

Define

$$C_* := \frac{N}{\sqrt{2}}, \qquad \delta_* := C_* - 1 > 0, \tag{35}$$

where $\delta_* > 0$ because $N \geq 2$, and

$$\boxed{\Lambda := \min\left\{1, G^{-(1+\alpha)}, \left(\frac{\delta_*}{2k_1 B_0^{r_1}}\right)^{\frac{1+\alpha}{r_1-1}}, \left(\frac{\delta_*}{2k_2 G^{r_2}}\right)^{\frac{1+\alpha}{r_2-1}}\right\}.} \tag{36}$$

Since $r_1, r_2 > 1$, one has $\Lambda > 0$.

For $0 < \lambda \leq \Lambda$ set

$$\underline{u}_\lambda := u_{0,\lambda}, \qquad \overline{u}_\lambda := t_\lambda \vartheta. \tag{37}$$

Theorem 3.1. *Assume $0 < \alpha < 1$ and (2) with constant exponents $r_1, r_2 > 1$. For every $\lambda \in (0, \Lambda]$, $\underline{u}_\lambda$ is a subsolution of (3), $\overline{u}_\lambda$ is a supersolution of (3), and*

$$0 < \underline{u}_\lambda \leq \overline{u}_\lambda \quad \text{in } \Omega.$$

Moreover,

$$\|\overline{u}_\lambda\|_{C^1(\overline{\Omega})} \leq (B_0 + G)\lambda^{\frac{1}{1+\alpha}}, \qquad \|\nabla \overline{u}_\lambda\|_\infty \leq G t_\lambda \leq 1. \tag{38}$$

Proof. Since $u_{0,\lambda}$ solves (32) and $f \geq 0$, it is immediately a subsolution of (3).

Set $t = t_\lambda$. By (36), $t \leq G^{-1}$, whence $|\nabla(t\vartheta)| \leq Gt \leq 1$. Thus the coefficient a coincides along $t\vartheta$ with the mean-curvature coefficient. Writing $y = x - x_0$, direct calculation gives

$$\begin{aligned} &-\operatorname{div}\left(\frac{\nabla(t\vartheta)}{\sqrt{1+|\nabla(t\vartheta)|^2}}\right) \\ &= \frac{2Nt}{\sqrt{1+4t^2|y|^2}} - \frac{8t^3|y|^2}{(1+4t^2|y|^2)^{3/2}} \\ &= \frac{2t\,[N + 4(N-1)t^2|y|^2]}{(1+4t^2|y|^2)^{3/2}}. \end{aligned} \tag{39}$$

Consequently,

$$-\operatorname{div}\left(a(|\nabla \overline{u}_\lambda|^2)\nabla \overline{u}_\lambda\right) \geq \frac{2Nt}{(1+4t^2R_\Omega^2)^{3/2}} \geq \frac{N}{\sqrt{2}}t = C_* t. \tag{40}$$

Since $\lambda = t^{1+\alpha}$ and $\vartheta \geq 1$,

$$\frac{\lambda}{\overline{u}_\lambda^\alpha} = t\vartheta^{-\alpha} \leq t. \tag{41}$$

Also, by (2) and (34),

$$f(x, \overline{u}_\lambda, \nabla \overline{u}_\lambda) \leq k_1 B_0^{r_1} t^{r_1} + k_2 G^{r_2} t^{r_2}. \tag{42}$$

The definition of Λ yields

$$k_1 B_0^{r_1} t^{r_1} \leq \frac{\delta_*}{2}t, \qquad k_2 G^{r_2} t^{r_2} \leq \frac{\delta_*}{2}t.$$

Hence

$$\frac{\lambda}{\overline{u}_\lambda^\alpha} + f(x, \overline{u}_\lambda, \nabla \overline{u}_\lambda) \leq (1+\delta_*)t = C_* t.$$

Together with (40), this proves that $\overline{u}_\lambda$ is a supersolution of (3).

Since it is also a supersolution of the singular problem (32), while $u_{0,\lambda}$ is a solution of (32), Lemma 2.9 (applied with $u_1 = u_{0,\lambda}$, $u_2 = \overline{u}_\lambda$; note that $(u_{0,\lambda} - \overline{u}_\lambda)^+$ has compact support in Ω, since $\overline{u}_\lambda \geq t_\lambda > 0$ on $\overline{\Omega}$) gives

$$u_{0,\lambda} \leq \overline{u}_\lambda \quad \text{in } \Omega,$$

because $u_{0,\lambda} = 0 \leq \overline{u}_\lambda$ on $\partial\Omega$. Thus the pair is ordered. Finally (34) gives (38). □

4. Gradient contraction and truncated problem

The argument below follows the contraction–truncation strategy used for convection terms in [7]. The essential point is that the gradient dependence is first contracted to a bounded ball and only then the order truncation associated with the sub–supersolution pair is introduced. This separates existence from the a priori estimate needed to remove the contraction.

Put

$$t_\lambda := \lambda^{1/(1+\alpha)}.$$

For $R > 0$ define the radial retraction $\pi_R : \mathbb{R}^N \to \overline{B}_R(0)$ by

$$\pi_R(\xi) = \begin{cases} \xi, & |\xi| \le R, \\ R\,\dfrac{\xi}{|\xi|}, & |\xi| > R, \end{cases}$$

and set

$$f_R(x, s, \xi) := f(x, s, \pi_R(\xi)).$$

Let

$$T_\lambda(s)(x) := \begin{cases} \underline{u}_\lambda(x), & s < \underline{u}_\lambda(x), \\ s, & \underline{u}_\lambda(x) \le s \le \overline{u}_\lambda(x), \\ \overline{u}_\lambda(x), & s > \overline{u}_\lambda(x). \end{cases}$$

For $u \in H^1_0(\Omega)$ we write simply $T_\lambda u(x) = T_\lambda(u(x))(x)$.

We consider

$$\begin{cases} -\operatorname{div}\left(a(|\nabla u|^2)\nabla u\right) &= \dfrac{\lambda}{(T_\lambda u)^\alpha} + f_R(x, T_\lambda u, \nabla(T_\lambda u)) \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega. \end{cases} \tag{43}$$

Lemma 4.1. *There exists $C_\lambda > 0$ such that*

$$\left| \int_\Omega (T_\lambda u)^{-\alpha} \varphi \, dx \right| \le C_\lambda \|\varphi\|_{H^1_0(\Omega)}$$

for all $u, \varphi \in H^1_0(\Omega)$.

Proof. Since $T_\lambda u \ge \underline{u}_\lambda$ and the Hopf estimate for $\underline{u}_\lambda$ gives $\underline{u}_\lambda(x) \ge c_\lambda d(x)$, we have $(T_\lambda u)^{-\alpha} \le c_\lambda^{-\alpha} d(x)^{-\alpha}$. Therefore

$$\int_\Omega (T_\lambda u)^{-\alpha} |\varphi| \le C_\lambda \int_\Omega d^{1-\alpha} \frac{|\varphi|}{d} \le C_\lambda \|d^{1-\alpha}\|_2 \left\| \frac{\varphi}{d} \right\|_2.$$

Hardy's inequality completes the proof. □

Proposition 4.2. *For every sufficiently small $\lambda > 0$ and every $R > 0$, problem* (43) *possesses a weak solution $u_{\lambda,R} \in H^1_0(\Omega)$.*

Proof. Define $\mathcal{A}_{\lambda,R} : H^1_0(\Omega) \to H^{-1}(\Omega)$ by

$$\begin{aligned} \langle \mathcal{A}_{\lambda,R}(u), v \rangle = & \int_\Omega a(|\nabla u|^2)\nabla u \cdot \nabla v \, dx \\ & - \int_\Omega \left[\frac{\lambda}{(T_\lambda u)^\alpha} + f_R(x, T_\lambda u, \nabla(T_\lambda u)) \right] v \, dx. \end{aligned}$$

The principal part is bounded, coercive, strictly monotone and of type (S_+). By Lemma 4.1, the singular term is bounded in $H^{-1}(\Omega)$. Moreover, since $0 \le T_\lambda u \le \|\overline{u}_\lambda\|_\infty$ and $|\pi_R(\nabla T_\lambda u)| \le R$, assumption (2) gives

$$0 \le f_R(x, T_\lambda u, \nabla T_\lambda u) \le k_1 \|\overline{u}_\lambda\|_\infty^{r_1} + k_2 R^{r_2}.$$

Thus the lower-order part maps bounded sets into bounded subsets of $H^{-1}(\Omega)$.

Let $u_n \rightharpoonup u$ in $H_0^1(\Omega)$ with $\limsup\langle \mathcal{A}_{\lambda,R}(u_n), u_n - u\rangle \le 0$. By the compact embedding $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$, $u_n \to u$ in $L^2(\Omega)$. Since the contracted convection term is bounded in $L^\infty(\Omega)$ uniformly in n,

$$\left| \int_\Omega f_R(x, T_\lambda u_n, \nabla(T_\lambda u_n))(u_n - u)\, dx \right| \le C\|u_n - u\|_1 \to 0.$$

For the singular term, let $\delta > 0$ and split Ω into $\{d < \delta\}$ and $\{d \ge \delta\}$. Using $(T_\lambda u_n)^{-\alpha} \le c_\lambda^{-\alpha} d^{-\alpha}$ and Hardy's inequality as in Lemma 4.1,

$$\int_\Omega (T_\lambda u_n)^{-\alpha} |u_n - u|\, dx \le C\delta^{1-\alpha}\|\nabla(u_n - u)\|_2 + C\delta^{-\alpha}\|u_n - u\|_1.$$

Since (u_n) is bounded in $H_0^1(\Omega)$, letting first $n \to \infty$ and then $\delta \to 0$ shows that this integral tends to zero. Therefore

$$\limsup_{n\to\infty} \int_\Omega a(|\nabla u_n|^2)\nabla u_n \cdot (\nabla u_n - \nabla u)\, dx \le 0.$$

The (S_+) property yields $u_n \to u$ strongly in $H_0^1(\Omega)$. Hence $\mathcal{A}_{\lambda,R}$ is pseudomonotone.

Finally, using $a \ge c > 0$, Poincaré's inequality and the boundedness established above,

$$\langle \mathcal{A}_{\lambda,R}(u), u\rangle \ge c\|\nabla u\|_2^2 - C_{\lambda,R}\|\nabla u\|_2,$$

so $\mathcal{A}_{\lambda,R}$ is coercive. Surjectivity of bounded coercive pseudomonotone operators gives $\mathcal{A}_{\lambda,R}(u_{\lambda,R}) = 0$. □

Proposition 4.3. *Let $0 < \lambda \le \Lambda$ and $R \ge Gt_\lambda$. Then every solution supplied by Proposition 4.2 satisfies*

$$\underline{u}_\lambda \le u_{\lambda,R} \le \overline{u}_\lambda \qquad \text{a.e. in } \Omega.$$

Consequently $T_\lambda u_{\lambda,R} = u_{\lambda,R}$ and $u_{\lambda,R}$ solves

$$-\operatorname{div}\big(a(|\nabla u|^2)\nabla u\big) = \lambda u^{-\alpha} + f_R(x, u, \nabla u). \tag{44}$$

Proof. Set $u = u_{\lambda,R}$. Test the equation with $(u - \overline{u}_\lambda)^+$. On the set $\{u > \overline{u}_\lambda\}$ one has $T_\lambda u = \overline{u}_\lambda$ and $\nabla(T_\lambda u) = \nabla\overline{u}_\lambda$ a.e. Since $|\nabla\overline{u}_\lambda| \le Gt_\lambda \le R$ by (38), we have $\pi_R(\nabla\overline{u}_\lambda) = \nabla\overline{u}_\lambda$, so that $f_R(x, T_\lambda u, \nabla(T_\lambda u)) = f(x, \overline{u}_\lambda, \nabla\overline{u}_\lambda)$ on this set. (This is the only place where the restriction $R \ge Gt_\lambda$ is used: no monotonicity of f in ξ is assumed.) Subtracting the supersolution inequality for $\overline{u}_\lambda$ gives

$$\int_{\{u>\overline{u}_\lambda\}} \big[a(|\nabla u|^2)\nabla u - a(|\nabla\overline{u}_\lambda|^2)\nabla\overline{u}_\lambda\big] \cdot (\nabla u - \nabla\overline{u}_\lambda)\, dx \le 0.$$

By (19), $(u - \overline{u}_\lambda)^+ = 0$. Testing similarly with $(\underline{u}_\lambda - u)^+$, using the subsolution inequality and $f_R \ge 0$, gives $\underline{u}_\lambda \le u$. Hence $T_\lambda u = u$, and (44) follows. □

5. Removal of the gradient contraction

We record the uniform estimate needed to pass from (44) to the original equation. It is the singular analogue, for the present uniformly elliptic mean-curvature perturbation, of the a priori gradient estimate used in the contraction argument of [7].

We next record the two boundary estimates needed in the rescaled argument. They isolate the only delicate regularity point caused by the singular term. The second lemma uses precisely the singular boundary regularity result in [17, Theorem B.1], which is an adaptation of Lieberman's boundary regularity method [20] to right-hand sides with a boundary singularity.

Lemma 5.1 (Uniform linear boundary estimate)**.** *Let $\Omega \subset \mathbb{R}^N$ be a bounded C^2 domain and $0 < \alpha < 1$. Let $t_\lambda \to 0$ and define*

$$\mathcal{A}_\lambda(\xi) := a(t_\lambda^2|\xi|^2)\xi.$$

Assume that there exist constants $\nu, L > 0$, independent of λ, such that

$$\nu|\eta|^2 \le D\mathcal{A}_\lambda(\xi)\eta \cdot \eta, \qquad |D\mathcal{A}_\lambda(\xi)| \le L \tag{45}$$

for every $\xi, \eta \in \mathbb{R}^N$. Suppose that $w_\lambda \in H_0^1(\Omega) \cap L^\infty(\Omega)$ is positive and satisfies

$$-\operatorname{div} \mathcal{A}_\lambda(\nabla w_\lambda) = w_\lambda^{-\alpha} + h_\lambda(x) \quad \text{in } \Omega, \qquad w_\lambda = 0 \quad \text{on } \partial\Omega, \tag{46}$$

where, uniformly in λ,

$$0 \le h_\lambda \le H, \qquad \|w_\lambda\|_\infty \le M, \qquad w_\lambda(x) \ge c_0 d(x). \tag{47}$$

Then there exists $C_d > 0$, independent of λ, such that

$$c_0 d(x) \le w_\lambda(x) \le C_d d(x) \qquad (x \in \Omega). \tag{48}$$

Proof. Only the upper estimate has to be proved. Since $\partial\Omega$ is C^2, there exists $\rho_0 > 0$ such that

$$\Omega_{\rho_0} := \{x \in \Omega : 0 < d(x) < \rho_0\}$$

is a tubular neighborhood of the boundary, $d \in C^2(\overline{\Omega_{\rho_0}})$, $|\nabla d| = 1$, and $|D^2 d| \le K_d$ there.

Fix positive constants A, E and put

$$q(s) := As - Es^{2-\alpha}, \qquad \Psi(x) := q(d(x)).$$

Then

$$q'(s) = A - E(2-\alpha)s^{1-\alpha}, \qquad q''(s) = -E(2-\alpha)(1-\alpha)s^{-\alpha},$$

and

$$D^2\Psi = q''(d)\nabla d \otimes \nabla d + q'(d) D^2 d.$$

Since $\mathcal{A}_\lambda$ is independent of x,

$$\begin{aligned} -\operatorname{div} \mathcal{A}_\lambda(\nabla\Psi) &= -D\mathcal{A}_\lambda(\nabla\Psi) : D^2\Psi \\ &\ge \nu E(2-\alpha)(1-\alpha)d^{-\alpha} - C_1|q'(d)|, \end{aligned} \tag{49}$$

where C_1 depends only on N, L and the C^2 geometry of $\partial\Omega$.

We now fix the parameters. Set

$$c_\alpha := (2-\alpha)(1-\alpha) > 0.$$

First choose $K > 0$, depending only on the structural and geometric constants, so large that

$$\nu K c_\alpha \ge 4C_1 + 4. \tag{50}$$

Next choose $0 < \rho \le \rho_0$ sufficiently small that

$$K(2-\alpha)\rho \le \frac{1}{2}, \qquad K\rho \le \frac{1}{2}, \tag{51}$$

and, in addition,

$$\frac{\nu K c_\alpha}{4}\frac{2M}{\rho} \ge H \quad \text{and} \quad \frac{\nu K c_\alpha}{4}\frac{2M}{\rho}\rho^\alpha \ge \left(\frac{M}{\rho}\right)^{-\alpha}. \tag{52}$$

The last two requirements can indeed be achieved by decreasing ρ: the first left-hand side is of order ρ^{-1}, whereas in the second inequality the left-hand side is of order $\rho^{\alpha-1}$ and the right-hand side is of order ρ^α. Having fixed K and ρ, define

$$A := \frac{2M}{\rho}, \qquad E := KA\rho^\alpha. \tag{53}$$

All these constants are independent of λ.

For $0 \le s \le \rho$, we have

$$E(2-\alpha)s^{1-\alpha} \le KA\rho^\alpha(2-\alpha)\rho^{1-\alpha} = KA(2-\alpha)\rho \le \frac{A}{2}.$$

Consequently,

$$\frac{A}{2} \le q'(s) \le A, \qquad 0 \le s \le \rho, \tag{54}$$

and hence

$$\frac{A}{2}d \le \Psi \le Ad \qquad \text{in } \Omega_\rho. \tag{55}$$

Moreover, on the inner boundary $\{d = \rho\}$,

$$\begin{aligned} \Psi &= A\rho - E\rho^{2-\alpha} \\ &= 2M(1 - K\rho) \ge M. \end{aligned} \tag{56}$$

Thus $\Psi \ge M$ on the inner boundary $\{d = \rho\}$.

It remains to verify the supersolution inequality. By (49), (53), and (54),

$$\begin{aligned} -\operatorname{div} \mathcal{A}_\lambda(\nabla\Psi) &\ge \nu K A c_\alpha \rho^\alpha d^{-\alpha} - C_1 A \\ &= \nu K A c_\alpha \left(\frac{\rho}{d}\right)^\alpha - C_1 A. \end{aligned} \tag{57}$$

Set $\mathcal{T} := \nu K A c_\alpha \rho^\alpha d^{-\alpha}$. Since $0 < d \le \rho$, we have $\mathcal{T} \ge \nu K A c_\alpha$. Hence, by (50), $\frac{1}{4}\mathcal{T} \ge C_1 A$, and, by the first condition in (52), $\frac{1}{4}\mathcal{T} \ge \frac{\nu K c_\alpha}{4}\frac{2M}{\rho} \ge H$. On the other hand, (55) gives

$$\Psi^{-\alpha} \le \left(\frac{A}{2}\right)^{-\alpha} d^{-\alpha}.$$

The second condition in (52) is exactly

$$\frac{\nu K A c_\alpha \rho^\alpha}{4} \ge \left(\frac{A}{2}\right)^{-\alpha},$$

and therefore

$$\frac{1}{4}\mathcal{T} = \frac{\nu K A c_\alpha \rho^\alpha}{4} d^{-\alpha} \ge \Psi^{-\alpha}.$$

Combining these estimates, $-\operatorname{div} \mathcal{A}_\lambda(\nabla\Psi) \ge \mathcal{T} - C_1 A \ge \frac{3}{4}\mathcal{T}$, which yields

$$-\operatorname{div} \mathcal{A}_\lambda(\nabla\Psi) \ge \Psi^{-\alpha} + H \qquad \text{in } \Omega_\rho, \tag{58}$$

uniformly in λ.

On $\partial\Omega$ both w_λ and Ψ vanish, while on $\{d = \rho\}$, (47) and (56) give $w_\lambda \le M \le \Psi$. Testing the difference between (46) and (58) with $(w_\lambda - \Psi)^+$ gives

$$\begin{aligned} &\int_{\{w_\lambda > \Psi\}} \big(\mathcal{A}_\lambda(\nabla w_\lambda) - \mathcal{A}_\lambda(\nabla\Psi)\big) \cdot \nabla(w_\lambda - \Psi)\, dx \\ &\qquad \le \int_{\{w_\lambda > \Psi\}} (w_\lambda^{-\alpha} - \Psi^{-\alpha})(w_\lambda - \Psi)\, dx \le 0, \end{aligned}$$

because $s \mapsto s^{-\alpha}$ is decreasing. The strong monotonicity implied by (45) yields $w_\lambda \le \Psi$ in Ω_ρ. Therefore

$$w_\lambda \le Ad \qquad \text{in } \Omega_\rho.$$

If $d(x) \ge \rho$, then

$$w_\lambda(x) \le M \le \frac{M}{\rho} d(x).$$

Consequently

$$w_\lambda(x) \le \max\left\{A, \frac{M}{\rho}\right\} d(x) \qquad \text{in } \Omega,$$

which proves the assertion. □

Lemma 5.2 (Uniform singular $C^{1,\beta}$ estimate)**.** *Under the hypotheses of Lemma 5.1, there exist $\beta \in (0,1)$ and $C > 0$, independent of λ, such that*

$$\|w_\lambda\|_{C^{1,\beta}(\overline{\Omega})} \le C. \tag{59}$$

Consequently,

$$\|t_\lambda w_\lambda\|_{C^1(\overline{\Omega})} \le C t_\lambda. \tag{60}$$

Proof. By Lemma 5.1,

$$c_0 d(x) \le w_\lambda(x) \le C_d d(x).$$

In particular,

$$w_\lambda(x)^{-\alpha} \le c_0^{-\alpha} d(x)^{-\alpha}.$$

Since $0 \le h_\lambda \le H$, if $D = \operatorname{diam}\Omega$, then

$$\begin{aligned} 0 \le F_\lambda(x) &:= w_\lambda(x)^{-\alpha} + h_\lambda(x) \\ &\le c_0^{-\alpha} d(x)^{-\alpha} + H \le C_F d(x)^{-\alpha}, \end{aligned} \tag{61}$$

where

$$C_F := c_0^{-\alpha} + H D^\alpha$$

is independent of λ.

We now verify the structural hypotheses in [17, Theorem B.1]. From

$$\mathcal{A}_\lambda(\xi) = a(t_\lambda^2|\xi|^2)\xi$$

we have

$$D\mathcal{A}_\lambda(\xi) = a(t_\lambda^2|\xi|^2)I + 2t_\lambda^2 a'(t_\lambda^2|\xi|^2)\xi \otimes \xi. \tag{62}$$

By hypothesis (45) (which, for the coefficient (4), holds with $\nu = \nu_0$ and $L = 1$ by Remark 2.4), uniformly in λ,

$$\nu|\eta|^2 \le D\mathcal{A}_\lambda(\xi)\eta \cdot \eta, \qquad |D\mathcal{A}_\lambda(\xi)| \le L.$$

Thus the structural assumptions of [17, Theorem B.1] are satisfied in the uniformly elliptic case $p = 2$. Moreover, $\mathcal{A}_\lambda$ has no explicit x-dependence, so the corresponding continuity assumption in the spatial variable is automatic. The datum satisfies

$$0 \le F_\lambda(x) \le C_F d(x)^{-\delta}, \qquad \delta = \alpha \in (0,1),$$

by (61), and the additional boundary hypothesis required in that theorem,

$$0 \le w_\lambda(x) \le C_d d(x),$$

is exactly the upper estimate supplied by Lemma 5.1.

Therefore [17, Theorem B.1] yields an exponent $\beta \in (0,1)$ and a constant $C > 0$, depending only on the domain, the structural constants ν, L, α, C_F, and C_d (hence only on Ω, N, α, ν, L, c_0, M and H), such that

$$\|w_\lambda\|_{C^{1,\beta}(\overline{\Omega})} \le C.$$

Since all these quantities are uniform in λ, the same β and C work for the whole family. Finally,

$$\|t_\lambda w_\lambda\|_{C^1(\overline{\Omega})} \le t_\lambda \|w_\lambda\|_{C^1(\overline{\Omega})} \le C t_\lambda,$$

as claimed. □

Remark 5.3. *For the decontraction argument only a positive uniform Hölder exponent is needed. Accordingly, we use precisely the conclusion furnished by [17, Theorem B.1] and do not claim here that every $\beta < 1 - \alpha$ is admissible.*

Lemma 5.4 (Uniform lower barrier)**.** *There exists $c_1 > 0$, independent of $\lambda > 0$, such that*

$$u_{0,\lambda}(x) \ge c_1 t_\lambda d(x) \qquad \text{in } \Omega.$$

Proof. Let $v_\lambda := u_{0,\lambda}/t_\lambda$ and $\mathcal{A}_\lambda(\xi) := a(t_\lambda^2|\xi|^2)\xi$. Since $\lambda = t_\lambda^{1+\alpha}$, v_λ solves

$$-\operatorname{div}\mathcal{A}_\lambda(\nabla v_\lambda) = v_\lambda^{-\alpha} \quad \text{in } \Omega, \qquad v_\lambda = 0 \quad \text{on } \partial\Omega.$$

Let $\varphi_1 \in C^2(\overline{\Omega})$ be the positive first Dirichlet eigenfunction of $-\Delta$, normalized by $\|\varphi_1\|_\infty = 1$; by the Hopf lemma, $\varphi_1 \geq k\,d$ for some $k > 0$. For $\varepsilon > 0$ put $z := \varepsilon\varphi_1$. By Remark 2.4, $\|D\mathcal{A}_\lambda\| \leq 1$, hence

$$-\operatorname{div}\mathcal{A}_\lambda(\nabla z) = -D\mathcal{A}_\lambda(\nabla z) : D^2 z \leq N\varepsilon\|D^2\varphi_1\|_\infty.$$

Choose $\varepsilon_0 > 0$, depending only on N, α and Ω, such that $N\varepsilon_0^{1+\alpha}\|D^2\varphi_1\|_\infty \leq 1$. Then, since $z \leq \varepsilon_0$,

$$-\operatorname{div}\mathcal{A}_\lambda(\nabla z) \leq \varepsilon_0^{-\alpha} \leq z^{-\alpha} \quad \text{in } \Omega$$

for $z = \varepsilon_0\varphi_1$. Testing the difference of the two problems with $(z - v_\lambda)^+ \in H_0^1(\Omega)$ and using that $s \mapsto s^{-\alpha}$ is decreasing and (19) (which holds for $\mathcal{A}_\lambda$), we get $(z - v_\lambda)^+ = 0$. Hence $v_\lambda \geq \varepsilon_0\varphi_1 \geq \varepsilon_0 k\,d$, and $c_1 := \varepsilon_0 k$ does not depend on λ. □

Lemma 5.5. *There exist $\lambda_0 \in (0, \Lambda]$, $\beta \in (0,1)$ and $C_0 \geq G$, independent of $\lambda \in (0, \lambda_0]$, such that the following holds. Let $R_\lambda := C_0 t_\lambda$ and let u_{λ,R_λ} be any solution of* (43) *with $R = R_\lambda$ supplied by Proposition 4.2. Then u_{λ,R_λ} solves* (44)*,*

$$\|u_{\lambda,R_\lambda}\|_{C^{1,\beta}(\overline{\Omega})} \leq C_0 t_\lambda \quad \textit{and, in particular,} \quad \|u_{\lambda,R_\lambda}\|_{C^1(\overline{\Omega})} \leq C_0 t_\lambda.$$

Proof. The constants are fixed in the following order, so that no constant depends on one chosen later.

Step 1: structure. By Remark 2.4, the family $\mathcal{A}_\lambda(\xi) := a(t_\lambda^2|\xi|^2)\xi$ satisfies (45) with $\nu = \nu_0$ and $L = 1$, for every λ.

Step 2: barriers. Put $c_2 := B_0$ and let c_1 be given by Lemma 5.4. Both depend only on Ω, N, α (and on the fixed point x_0).

Step 3: the a priori constant. Apply Lemmas 5.1 and 5.2 with $\nu = \nu_0$, $L = 1$, $c_0 = c_1$, $M = c_2$ and $H = 1$. They give $\beta \in (0,1)$ and $C_\sharp > 0$, depending only on these quantities, on α and on Ω, such that every positive $w \in H_0^1(\Omega) \cap L^\infty(\Omega)$ satisfying (46) with

$$0 \leq h_\lambda \leq 1, \qquad c_1 d \leq w \leq c_2, \tag{63}$$

obeys $\|w\|_{C^{1,\beta}(\overline{\Omega})} \leq C_\sharp$. Define

$$C_0 := \max\{C_\sharp, G\}.$$

Step 4: smallness of λ. Since $r_1, r_2 > 1$, we can choose $\lambda_0 \in (0, \Lambda]$ such that, for $0 < \lambda \leq \lambda_0$,

$$k_1 c_2^{r_1} t_\lambda^{r_1 - 1} + k_2 C_0^{r_2} t_\lambda^{r_2 - 1} \leq 1 \qquad \text{and} \qquad C_0 t_\lambda < 1. \tag{64}$$

Step 5: conclusion. Let $0 < \lambda \leq \lambda_0$ and $u = u_{\lambda,R_\lambda}$. Since $R_\lambda = C_0 t_\lambda \geq G t_\lambda$, Proposition 4.3 applies: u solves (44) and $\underline{u}_\lambda \leq u \leq \overline{u}_\lambda$. Write $u = t_\lambda w$. Since $\lambda = t_\lambda^{1+\alpha}$, equation (44) becomes

$$-\operatorname{div}\big(a(t_\lambda^2|\nabla w|^2)\nabla w\big) = w^{-\alpha} + h_\lambda, \qquad h_\lambda := t_\lambda^{-1} f_{R_\lambda}(x, t_\lambda w, t_\lambda \nabla w). \tag{65}$$

By Lemma 5.4 and $\overline{u}_\lambda = t_\lambda \vartheta \leq t_\lambda B_0$, we have $c_1 d \leq w \leq c_2$. Moreover, since $|\pi_{R_\lambda}(\xi)| \leq C_0 t_\lambda$, assumption (2) and (64) give

$$0 \leq h_\lambda \leq t_\lambda^{-1}\big(k_1(t_\lambda c_2)^{r_1} + k_2(C_0 t_\lambda)^{r_2}\big) = k_1 c_2^{r_1} t_\lambda^{r_1-1} + k_2 C_0^{r_2} t_\lambda^{r_2-1} \leq 1.$$

Thus (63) holds and Step 3 yields $\|w\|_{C^{1,\beta}(\overline{\Omega})} \leq C_\sharp \leq C_0$. Since $u = t_\lambda w$,

$$\|u\|_{C^{1,\beta}(\overline{\Omega})} \leq C_\sharp t_\lambda \leq C_0 t_\lambda,$$

as claimed. □

Proposition 5.6. *For $0 < \lambda \leq \lambda_0$, every solution u_{λ,R_λ} as in Lemma 5.5 satisfies*

$$|\nabla u_{\lambda,R_\lambda}(x)| \leq R_\lambda \qquad \textit{for every } x \in \overline{\Omega}.$$

Hence

$$f_{R_\lambda}(x, u_{\lambda,R_\lambda}, \nabla u_{\lambda,R_\lambda}) = f(x, u_{\lambda,R_\lambda}, \nabla u_{\lambda,R_\lambda}),$$

and u_{λ,R_λ} solves the uncontracted auxiliary problem (3)*.*

Proof. This is immediate from Lemma 5.5 and the definition $R_\lambda = C_0 t_\lambda$. □

6. Proof of Theorem 1.1

Proof.

Let $\lambda^* := \lambda_0$, where $\lambda_0 \leq \Lambda$ is given by Lemma 5.5; in particular $C_0 t_\lambda < 1$ for $0 < \lambda \leq \lambda^*$, by (64). Let $u_\lambda = u_{\lambda,R_\lambda}$, which exists by Proposition 4.2. By Proposition 5.6, u_λ solves (3). Since $\|\nabla u_\lambda\|_\infty < 1$, the definition (4) gives

$$a(|\nabla u_\lambda|^2) = \frac{1}{\sqrt{1+|\nabla u_\lambda|^2}} \qquad \text{in } \overline{\Omega}.$$

Therefore (3) is exactly the original prescribed mean-curvature problem (1). Positivity follows from $u_\lambda \geq \underline{u}_\lambda > 0$ in Ω, and Lemma 5.5 yields

$$\|u_\lambda\|_{C^1(\overline{\Omega})} \leq C_0 \lambda^{1/(1+\alpha)} \longrightarrow 0 \qquad \text{as } \lambda \to 0^+.$$

Finally, Lemma 5.5 gives $u_\lambda \in C^{1,\beta}(\overline{\Omega})$ with $\beta \in (0,1)$ independent of λ. This proves Theorem 1.1. □

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Declarations

Conflict of interests. The authors have no competing interests to declare that are relevant to the content of this article.

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